

Yizhuo Wang

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Ph.D. Candidate, Multi-Agent Robotic Motion (MARMot) Laboratory, National University of Singapore.
Research interest: robot learning, path planning, vision-and-language navigation, reinforcement learning.

Education

- **Ph.D., Robotics, National University of Singapore** **Singapore**
Advised by Asst. Prof. Guillaume Sartoretti *Fall 2022 – Fall 2026*
 - Research topic: Machine learning based methods for scalable, long-term planning in large multi-agent teams.
- **Master, Mechanical Engineering (Control), National University of Singapore** **Singapore**
Advised by Asst. Prof. Guillaume Sartoretti, GPA: 4.5/5.0 *2020 – 2021*
 - Thesis title: Reinforcement Learning for Multi-Agent Search and Rescue.
- **Bachelor, Mechatronics, Harbin Institute of Technology** **Harbin & Suzhou, China**
Advised by Assoc. Prof. Ong Chong Jin *2016 – 2020*
 - Thesis title: Development of mini-AGV for cooperative control.

Work Experience

- **National University of Singapore** **Singapore**
Research Engineer *2021 – 2022*
 - Investigate advanced multi-agent planning/exploration methods, funded by Temasek Laboratories @ NUS.
- **Meet Hive Intelligence Co., Ltd.** **Xi'an, China**
Mechanical Development Intern *Summer 2020*
 - Designed the mechanical structure and visual detection system for Garbage Automatic Recycling System for Household (GARSH). Six patents attained.

Publications

Conference / Journal / Workshop

- Y. Cao, **Y. Wang**, J. Liang, Y. Zhang, G. Sartoretti. HEADER: Hierarchical Robot Exploration Attention Deep Networks with Expert-Guided Reinforcement Learning. **Submitted to IEEE Transactions on Robotics.** **T-RO 2026**
- R. Zhao, X. Zhang, Y. Cao, **Y. Wang**, and G. Sartoretti. Attention-based Learning for 3D Informative Path Planning. **Submitted to IEEE International Conference on Robotics and Automation**, Vienna, Austria. **ICRA 2026**
- X. Zhang, **Y. Wang**, G. Sartoretti. COMPASS: Cooperative Multi-Agent Persistent Monitoring using Spatio-Temporal Attention Network. In *IEEE International Symposium on Multi-Robot & Multi-Agent Systems*, Singapore. **MRS 2025**
- Y. Cao, **Y. Wang**, J. Liang, G. Sartoretti. Scalable and Expert-guided Reinforcement Learning-based Autonomous Robot Exploration. In *Active Perception Workshop*. **IROS 2025**
- **Y. Wang**, H. He, J. Liang, Y. Cao, R. Chakraborty, and G. Sartoretti. CogniPlan: Uncertainty-Guided Path Planning with Conditional Generative Layout Prediction. In *Conference on Robot Learning*, Seoul, Korea. [\[Project Page\]](#) **CoRL 2025**

- J. Chiun, S. Zhang, **Y. Wang**, Y. Cao, G. Sartoretti. MARVEL: Multi-Agent Reinforcement Learning for constrained field-of-View multi-robot Exploration in Large-scale environments. In *IEEE International Conference on Robotics and Automation*, Atlanta, USA. **ICRA 2025**
- **Y. Wang**, Y. Cao, J. Chiun, S. Koley, M. Pham, G. Sartoretti. ViPER: Visibility-based Pursuit-Evasion via Reinforcement Learning. In *Conference on Robot Learning*, Munich, Germany. **CoRL 2024**
- Y. Cao, R. Zhao, **Y. Wang**, B. Xiang, G. Sartoretti. Deep Reinforcement Learning-based Large-scale Robot Exploration. In *IEEE Robotics and Automation Letters* (vol.9, no.5, pp.4631-4638). **RA-L 2024**
- **Y. Wang**, Y. Wang, Y. Cao, G. Sartoretti. Spatio-Temporal Attention Network for Persistent Monitoring of Multiple Mobile Targets. In *IEEE/RSJ International Conference on Intelligent Robots and Systems* (pp.3903-3910), Detroit, USA. **IROS 2023**
- Y. Wang, **Y. Wang**, G. Sartoretti. Full Communication Memory Networks for Team-Level Cooperation Learning. In Springer's *Autonomous Agents and Multi-Agent Systems*, 37, 33. **J-AAMAS 2023**
- **Y. Wang**, G. Sartoretti. Learning Simultaneous Motion Planning and Active Gaze Control for Persistent Monitoring of Dynamic Targets. In *Workshop on Active Methods in Autonomous Navigation*. **ICRA 2023**
- Y. Cao, T. Hou, **Y. Wang**, X. Yi, and G. Sartoretti. ARiADNE: A Reinforcement learning approach using Attention-based Deep Networks for Exploration. In *IEEE International Conference on Robotics and Automation* (pp.10219-10225), London, UK. **ICRA 2023**
- Y. Cao, **Y. Wang**, A. Vashisth, H. Fan, and G. Sartoretti. CATNIPP: Context-Aware Attention-based Network for Informative Path Planning. In *Conference on Robot Learning* (pp.1928-1937), Auckland, New Zealand. PMLR. **CoRL 2022**

Selected Patent.....

- Z. Luo, **Y. Wang**, Y. Fu, J. Lu, and H. Zhang, A Rotating Structure for Garbage Sorting. *Utility Model Patent (Chinese)*, CN213474287U. **2021**
- Z. Luo, **Y. Wang**, Y. Fu, J. Lu, and H. Zhang, A Garbage Pushing and Delivery Structure. *Utility Model Patent (Chinese)*, CN213325514U. **2021**
- **Y. Wang**, Y. Fu, Z. Luo, H. Zhang, and J. Lu. Intelligent Recycling Bin (Dys-1). *Design Patent (Chinese)*, CN306247881S. **2020**

Profession Activities

Teaching.....

- **Teaching Assistant** **Singapore**
- *NUS ME3103 Mechanical Systems Design* *Spring 2021*

Competition.....

- **Singapore Amazing Flying Machine Competition 2023** **Singapore**
- *Judge Commendation Award & Championship Award – 5th place* *2023*
- **6th China Intl. College Students 'Internet+' Innov. & Entrep. Competition** **Guangzhou, China**
- *Gold Prize (Red Tour Track), Nationwide* *2020*

Paper Review.....

- *IEEE Robotics and Automation Letters (RA-L)* *2023/2024/2025*
- *IEEE International Conference on Robotics and Automation (ICRA)* *2025*
- *Robotics: Science and Systems (RSS)* *2024*
- *Association for the Advancement of Artificial Intelligence (AAAI)* *2023/2024/2025*
- *Conference on Robot Learning (CoRL)* *2023/2024/2025*

- European Conference on Artificial Intelligence (ECAI) 2023
- IEEE Transactions on Automation Science and Engineering (T-ASE) 2023
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2023/2024/2025
- International Symposium on Experimental Robotics (ISER) 2023
- International Joint Conference on Artificial Intelligence (IJCAI) 2022/2023